

Automatic Restricted Quantity LIQUID DISPENSER

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This automatic restricted quantity water or petrol dispenser is based on RFID reader and Arduino. Whenever you want to fill a water bottle or can, or petrol in a vehicle, or any other liquid for that matter, you just need to place the correct RFID card for the quantity of liquid required. The Arduino board reads the data from the RFID reader, which in turn drives the relays through ULN IC for that particular quantity.

This system helps in marriages and other functions, where you see that even after filling water from water can, if children or anyone leaves the tap open, then the water keeps flowing and gets wasted. Water saving will increase with this project. It also provides the security for the customers from various frauds that occur at petrol pumps, like short fueling of petrol by distraction, tweaking petrol-dispensing

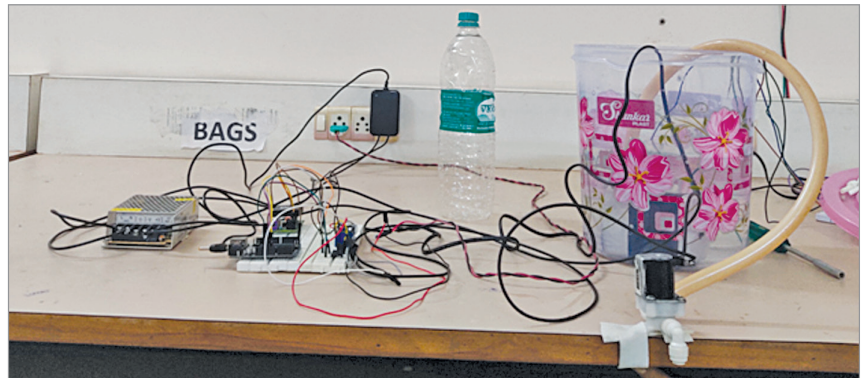


Fig. 1: Author's prototype

machines using integrated chips.

In this proposed project, the quantity of liquid dispensed de-

pends on the delay time for each digital output pin of the Arduino, which gets added through Arduino programming. Here we have used five different RFID tags for the different delay times required to dispense 1/2 litre, 1 litre, 1.5 litre, 2 litres, and 2.5 litres. A solenoid valve has been used for the liquid to flow without much force.

The author's prototype is shown in Fig. 1. The components required for the project are mentioned under the Bill of Material table.

TABLE 1
BILL OF MATERIAL

Components	Quantity
Arduino Uno (MOD1)	1
EM-18 RFID reader module (MOD2)	1
RFID tags/cards (125kHz)	4
ULN 2003 IC (IC1)	1
Water pump, 220V AC (Pump1)	1
Solenoid valve, 24V DC	1
12V DC power adaptor	1
Relay, 5V DC SPST (R1, R2)	2

```
//RAKESH JAIN PROGRAM TO READ RFID TAG
void setup() {
  Serial.begin(9600);
}

void loop() {
  if(Serial.available())
  {
    Serial.print((char)Serial.read());
  }
}
```

Fig. 2: Code to obtain RFID card ID number

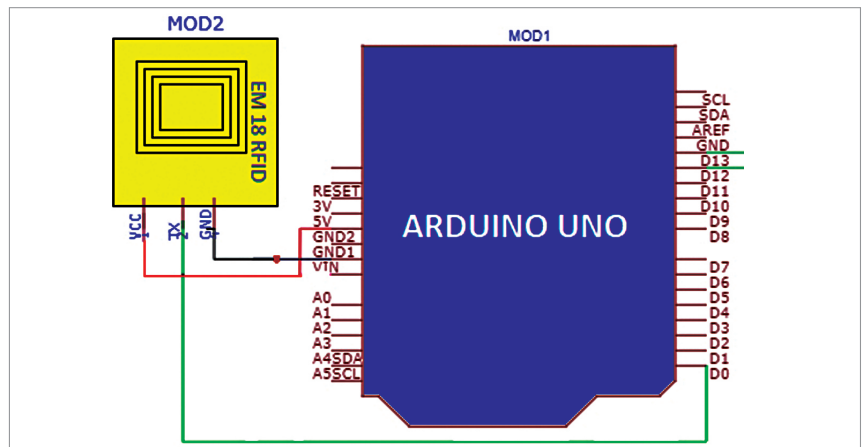


Fig. 3: EM-18 and Arduino programing circuit

EFY Note

The source code of this project is available for free download at www.electronicsforu.com